

14

The Friedmann–Robertson–Walker geometry

We now discuss the application of general relativity to modelling the behaviour of the universe as a whole. In order to do this, we make some far-reaching assumptions, but only those consistent with our observations of the universe. As in our derivations of the Schwarzschild and Kerr geometries, we begin by using symmetry arguments to restrict the possible forms for the metric describing the overall spacetime geometry of the universe.¹

14.1 The cosmological principle

When we look up at the sky we see that the stars around us are grouped into a large-density concentration – the Milky Way Galaxy. On a slightly larger scale, we see that our Galaxy belongs to a small group of galaxies (called the Local Group). Our Galaxy and our nearest large neighbour, the Andromeda galaxy, dominate the mass of the Local Group. On still larger scales we see that our Local Group sits on the outskirts of a giant supercluster of galaxies centred in the constellation of Virgo. Evidently, on small scales matter is distributed in a highly irregular way but, as we look on larger and larger scales, the matter distribution looks more and more uniform. In fact, we have very good evidence (particularly from the constancy of the temperature of the cosmic microwave background in different directions on the sky) that the universe is *isotropic* on the very largest scales, to high accuracy. If the universe has no preferred centre then *isotropy* also implies *homogeneity*. We therefore have good physical reasons to study simple cosmological models in which the universe is assumed to be homogeneous and

¹ For a detailed discussion, see, for example, J. N. Islam, *An Introduction to Mathematical Cosmology*, Cambridge University Press, 1992.

isotropic². We thus assume the *cosmological principle*, which states that *at any particular time the universe looks the same from all positions in space at a particular time and all directions in space at any point are equivalent*.

14.2 Slicing and threading spacetime

The intuitive statement of the cosmological principle given above needs to be made more precise. In particular, how does one define a ‘particular time’ in general relativity that is valid globally, when there are no global inertial frames? Also, since observers moving relative to one another will view the universe differently, according to which observers do we demand the universe to appear isotropic?

In general relativity the concept of a ‘moment of time’ is ambiguous and is replaced by the notion of a *three-dimensional spacelike hypersurface*. To define a ‘time’ parameter that is valid globally, we ‘slice up’ spacetime by introducing a series of non-intersecting spacelike hypersurfaces that are labelled by some parameter t . This parameter then defines a universal time in that ‘a particular time’ means a given spacelike hypersurface. It should be noted, however, that we may construct the hypersurfaces $t = \text{constant}$ in any number of ways. In a general spacetime there is no preferred ‘slicing’ and hence no preferred ‘time’ coordinate t .

It is useful at this point to introduce the idealised concept of *fundamental observers*, who are assumed to have no motion relative to the overall *cosmological fluid* associated with the ‘smeared-out’ motion of all the galaxies and other matter in the universe. A fundamental observer would, for example, measure no dipole moment in his observations of the cosmic microwave background radiation; an observer with a non-zero *peculiar velocity* would observe such a dipole as a result of the Doppler effect arising from his motion relative to the cosmological fluid. Adopting *Weyl’s postulate*, the timelike worldlines of these observers are assumed to form a bundle, or *congruence*, in spacetime that diverges from a point in the (finite or infinitely distant) past or converges to such a point in the future. These worldlines are non-intersecting, except possibly at a singular point in the past or future or both. Thus, there is a unique worldline passing through each (non-singular) spacetime point. The set of worldlines is sometimes described as providing *threading* for the spacetime.

The hypersurfaces $t = \text{constant}$ may now be naturally constructed in such a way that the 4-velocity of any fundamental observer is orthogonal to the hypersurface.

² It is worth noting that isotropy about every point automatically implies homogeneity. However, homogeneity does *not* necessarily imply isotropy. For example, a universe with a large-scale magnetic field that pointed in one direction everywhere and had the same magnitude at every point would be homogeneous but not isotropic.

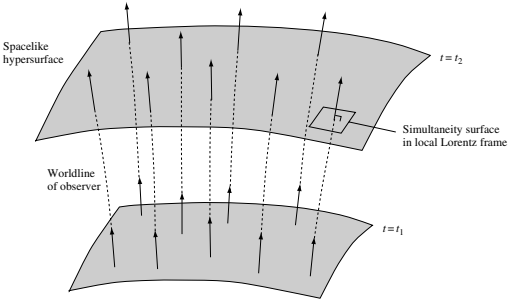


Figure 14.1 Representation (with one spatial dimension suppressed) of spacelike hypersurfaces on which fundamental observers are assumed to lie. The worldline of any fundamental observer is orthogonal to any such surface.

Thus, the surface of simultaneity of the local Lorentz frame of any such observer coincides locally with the hypersurface (see Figure 14.1). Each hypersurface may therefore be considered as the ‘meshing together’ of all the local Lorentz frames of the fundamental observers.

14.3 Synchronous coordinates

The spacelike hypersurfaces discussed above are labelled by a parameter t , which may be taken to be the *proper time* along the worldline of any fundamental observer. The parameter t is then called the *synchronous time coordinate*. In addition, we may also introduce spatial coordinates (x^1, x^2, x^3) that are *constant* along any worldline. Thus each fundamental observer has fixed (x^1, x^2, x^3) coordinates, and so the latter are called *comoving coordinates*. Since each hypersurface $t = \text{constant}$ is orthogonal to the observer’s worldline, the line element takes the form

$$ds^2 = c^2 dt^2 - g_{ij} dx^i dx^j \quad (\text{for } i, j = 1, 2, 3), \quad (14.1)$$

where the g_{ij} are functions of the coordinates (t, x^1, x^2, x^3) .

We may verify that the metric (14.1) does indeed incorporate the properties described in the previous section, as follows. Let $x^\mu(\tau)$ be the worldline of a

fundamental observer, where τ is the proper time along the worldline. Then, by construction, $x^\mu(\tau)$ is given by

$$x^0 = \tau, \quad x^1 = \text{constant}, \quad x^2 = \text{constant}, \quad x^3 = \text{constant}. \quad (14.2)$$

Since $dx^i = 0$ along the worldline, we obtain $ds = c d\tau = c dt$ and so $t = \tau$, thereby showing that the proper time τ along the worldline is indeed equal to t . Thus, from (14.2), it is clear that the 4-velocity of a fundamental observer in comoving coordinates is

$$[u^\mu] \equiv \left[\frac{dx^\mu}{d\tau} \right] = (1, 0, 0, 0). \quad (14.3)$$

Since any vector lying in the hypersurface $t = \text{constant}$ has the form $[a^\mu] = (0, a^1, a^2, a^3)$, we see that

$$g_{\mu\nu} u^\mu a^\nu = 0,$$

because $g_{0i} = 0$ for $i = 1, 2, 3$. Hence, the observer's 4-velocity is orthogonal to the hypersurface, as we required. Finally, we may show that the worldline given by (14.2) satisfies the geodesic equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\sigma} \frac{dx^\nu}{d\tau} \frac{dx^\sigma}{d\tau} = 0.$$

Using (14.3), we see that we require only that $\Gamma^\mu_{00} = 0$. This quantity is given by

$$\Gamma^\mu_{00} = \frac{1}{2} g^{\mu\nu} (2\partial_0 g_{0\nu} - \partial_\nu g_{00}),$$

which is easily shown to be zero by using the fact that $g^{0i} = 0$ for $i = 1, 2, 3$. Thus the worldlines $x^\mu(\tau)$ are geodesics and hence can describe particles (observers) moving only under the influence of gravity.

14.4 Homogeneity and isotropy of the universe

The metric (14.1) does *not* yet incorporate the property that space is homogeneous and isotropic. Indeed this form of the metric can be used, with the help of a special coordinate system obtained by singling out a particular fundamental observer, to derive some general properties of the universe, *without the assumptions of homogeneity and isotropy*, although we will not consider such cases here.

Let us now incorporate the postulates of homogeneity and isotropy. The former demands that all points on a particular spacelike hypersurface are equivalent, whereas the latter demands that all directions on the hypersurface are equivalent for fundamental observers. The (squared) spatial separation on the same

hypersurface $t = \text{constant}$ of two nearby galaxies at coordinates (x^1, x^2, x^3) and $(x^1 + \Delta x^1, x^2 + \Delta x^2, x^3 + \Delta x^3)$ is

$$d\sigma^2 = g_{ij}\Delta x^i\Delta x^j.$$

If we consider the triangle formed by three nearby galaxies at some particular time t , then isotropy requires that the triangle formed by these same galaxies at some later time must be similar to the original triangle. Moreover, homogeneity requires that the magnification factor must be *independent* of the position of the triangle in the 3-space. It thus follows that time t can enter the g_{ij} only through a common factor, so that the ratios of small distances are the same at all times. Hence the metric must take the form

$$\boxed{ds^2 = c^2 dt^2 - S^2(t)h_{ij}dx^i dx^j}, \quad (14.4)$$

where $S(t)$ is a time-dependent *scale factor* and the h_{ij} are functions of the coordinates (x^1, x^2, x^3) only. We note that it is common practice to identify fundamental observers loosely with individual galaxies (which are assumed to be pointlike). However, since the magnification factor is independent of position, we must neglect the small peculiar velocities of real individual galaxies.

14.5 The maximally symmetric 3-space

We clearly require the 3-space spanned by the spacelike coordinates (x^1, x^2, x^3) to be homogeneous and isotropic. This leads us to study the *maximally symmetric 3-space*. In three dimensions, the curvature tensor R_{ijkl} has, in general, six independent components, each of which is a function of the coordinates. We therefore need to specify six functions to define the intrinsic geometric properties of a general three-dimensional space. Clearly, the more symmetrical the space, the fewer the functions needed to specify its properties. A *maximally symmetric space* is specified by just one number – the *curvature* K , which is independent of the coordinates. Such *constant curvature* spaces must clearly be *homogeneous and isotropic*.

The curvature tensor of a maximally symmetric space must take a particularly simple form. It must clearly depend on the constant K and on the metric tensor g_{ij} . The simplest expression that satisfies the various symmetry properties and identities of R_{ijkl} and contains just K and the metric tensor is given by

$$R_{ijkl} = K(g_{ik}g_{jl} - g_{il}g_{jk}). \quad (14.5)$$

In fact, a maximally symmetric space is *defined* as one having a curvature tensor of the form (14.5).

The Ricci tensor is given by

$$\begin{aligned} R_{jk} &= g^{il} R_{ijkl} = K g^{il} (g_{ik} g_{jl} - g_{il} g_{jk}) \\ &= K (\delta_k^l g_{jl} - \delta_l^j g_{jk}) \\ &= K (g_{jk} - 3g_{jk}) = -2K g_{jk}. \end{aligned}$$

The curvature scalar is thus given by

$$R = R_k^k = -2K \delta_k^k = -6K.$$

As in our derivation of the general static isotropic metric in Section 9.1, the metric of an isotropic 3-space must depend only on the rotational invariants

$$\vec{x} \cdot \vec{x} \equiv r^2, \quad d\vec{x} \cdot d\vec{x}, \quad \vec{x} \cdot d\vec{x},$$

and in spherical polar coordinates (r, θ, ϕ) it must take the form

$$\begin{aligned} d\sigma^2 &= C(r)(\vec{x} \cdot d\vec{x})^2 + D(r)(d\vec{x} \cdot d\vec{x})^2 \\ &= C(r)r^2 dr^2 + D(r)(dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2). \end{aligned}$$

Following our analysis in Chapter 9, we can simplify this line element by redefining the radial coordinate $\tilde{r}^2 = r^2 D(r)$. Dropping the bars on the variables, the metric can thus be written as

$$d\sigma^2 = B(r) dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2,$$

where $B(r)$ is an arbitrary function of r .

We have met this line element before – it is identical to the space part of the general static isotropic metric. In Chapter 9, we showed that the only non-zero connection coefficients are

$$\begin{aligned} \Gamma_{rr}^r &= \frac{1}{2B(r)} \frac{dB(r)}{dr}, & \Gamma_{\theta\theta}^r &= -\frac{r}{B(r)}, & \Gamma_{\phi\phi}^r &= -\frac{r \sin^2 \theta}{B(r)}, \\ \Gamma_{r\theta}^\theta &= \Gamma_{r\phi}^\phi = \frac{1}{r}, & \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta, & \Gamma_{\phi\theta}^\phi &= \cot \theta. \end{aligned}$$

The Ricci tensor is given in terms of the connection coefficients by

$$R_{ij} = \partial_j \Gamma_{ik}^k - \partial_k \Gamma_{ij}^k + \Gamma_{ik}^l \Gamma_{lj}^k - \Gamma_{ij}^l \Gamma_{lk}^k,$$

and, after some algebra, we find that its non-zero components are

$$\begin{aligned} R_{rr} &= -\frac{1}{rB} \frac{dB}{dr}, \\ R_{\theta\theta} &= \frac{1}{B} - 1 - \frac{r}{2B^2} \frac{dB}{dr} \\ R_{\phi\phi} &= R_{\theta\theta} \sin^2 \theta \end{aligned}$$

For our 3-space to be maximally symmetric, however, we must have

$$R_{ij} = -2K g_{ij},$$

and so we require

$$\frac{1}{rB} \frac{dB}{dr} = 2KB(r), \quad (14.6)$$

$$1 + \frac{r}{2B^2} \frac{dB}{dr} - \frac{1}{B} = 2Kr^2. \quad (14.7)$$

Integrating (14.6) we immediately obtain

$$B(r) = \frac{1}{A - Kr^2},$$

where A is a constant of integration. Substituting this expression into (14.7) then gives

$$1 - A + Kr^2 = Kr^2,$$

from which we see that $A = 1$. Thus, we have constructed the line element for the maximally symmetric 3-space, which takes the form

$$d\sigma^2 = \frac{dr^2}{1 - Kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (14.8)$$

and has a curvature tensor specified by one number, K , the *curvature* of the space.

Notice also that this is *exactly* the same form as the metric for a 3-sphere embedded in four-dimensional Euclidean space, which we discussed in Chapter 2. The metric contains a ‘hidden symmetry’, since *the origin of the radial coordinate is completely arbitrary*. We can choose any point in this space as our origin since all points are equivalent. *There is no centre in this space*. We also note that, on scales small compared with the spatial curvature, the line element (14.8) is equivalent to that of a three-dimensional Euclidean space.

14.6 The Friedmann–Robertson–Walker metric

Combining the our expression (14.8) for the maximally symmetric 3-space with the line element (14.4), which incorporates the cosmological principle and Weyl's postulate, we obtain

$$ds^2 = c^2 dt^2 - S^2(t) \left[\frac{dr^2}{1 - Kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right]. \quad (14.9)$$

It is usual to write this line element in an alternative form in which the arbitrariness in the magnitude of K is absorbed into the radial coordinate and the scale factor. Assuming firstly that $K \neq 0$ we define the variable $k = K/|K|$ in such a way that $k = \pm 1$ depending on whether K is positive or negative. If we introduce the rescaled coordinate

$$\bar{r} = |K|^{1/2} r,$$

then (14.9) becomes

$$ds^2 = c^2 dt^2 - \frac{S^2(t)}{|K|} \left[\frac{d\bar{r}^2}{1 - k\bar{r}^2} + \bar{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right].$$

Finally, we define a rescaled scale function $R(t)$ by

$$R(t) = \begin{cases} \frac{S(t)}{|K|^{1/2}} & \text{if } K \neq 0, \\ S(t) & \text{if } K = 0. \end{cases}$$

Then, dropping the bars on the radial coordinate, we obtain the standard form for the *Friedmann–Robertson–Walker* (FRW) line element,

$$ds^2 = c^2 dt^2 - R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right], \quad (14.10)$$

where k takes the values -1 , 0 , or 1 depending on whether the spatial section has negative, zero or positive curvature respectively. It is also clear that the coordinates (r, θ, ϕ) appearing in the FRW metric are still *comoving*, i.e. the worldline of a galaxy, ignoring any peculiar velocity, has fixed values of (r, θ, ϕ) .

14.7 Geometric properties of the FRW metric

The geometric properties of the homogeneous and isotropic 3-space corresponding to the hypersurface $t = \text{constant}$ depend upon whether $k = -1$, 0 or 1 . We now consider each of these cases in turn.

Positive spatial curvature: $k = 1$

In the case $k = 1$, we see that the coefficient of dr in the FRW metric becomes singular as $r \rightarrow 1$. We therefore introduce a new radial coordinate χ , defined by the relation

$$r = \sin \chi \quad \Rightarrow \quad dr = \cos \chi d\chi = (1 - r^2)^{1/2} d\chi,$$

so that the spatial part of the FRW metric takes the form

$$d\sigma^2 = R^2 [d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2)],$$

where R is the value of the scale factor at the particular time t defining the spacelike hypersurface of interest.

Some insight into this spatial metric may be gained by considering the 3-space as *embedded* in a four-dimensional Euclidean space with coordinates (w, x, y, z) , where

$$\begin{aligned} w &= R \cos \chi, \\ x &= R \sin \chi \sin \theta \cos \phi, \\ y &= R \sin \chi \sin \theta \sin \phi, \\ z &= R \sin \chi \cos \theta. \end{aligned}$$

In fact we have already discussed exactly this embedding in Section 2.9. Such an embedding is possible since one can write

$$d\sigma^2 = dw^2 + dx^2 + dy^2 + dz^2 = R^2 [d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2)],$$

where, from the transformation equations, we have the constraint

$$w^2 + x^2 + y^2 + z^2 = R^2.$$

This shows that our 3-space can be considered as a three-dimensional sphere in the four-dimensional Euclidean space. This hypersurface is defined by the coordinate ranges

$$0 \leq \chi \leq \pi, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi \leq 2\pi.$$

The surfaces $\chi = \text{constant}$ are 2-spheres with surface area

$$A = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (R \sin \chi d\theta)(R \sin \chi \sin \theta d\phi) = 4\pi R^2 \sin^2 \chi,$$

and (θ, ϕ) are the standard spherical polar coordinates of these 2-spheres. Thus, as χ varies from 0 to π , the area of the 2-spheres increases from zero to a maximum value of $4\pi R^2$ at $\chi = \pi/2$, after which it decreases to zero at $\chi = \pi$. The proper

radius of a 2-sphere is $R\chi$, and so the surface area is smaller than that of a sphere of radius $R\chi$ in Euclidean space.

The entire 3-space has a *finite* total volume given by

$$V = \int_{\chi=0}^{\pi} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (R d\chi)(R \sin \chi d\theta)(R \sin \chi \sin \theta d\phi) = 2\pi^2 R^3,$$

which is the reason why, in this case, R is often referred to as the ‘radius of the universe’.

Zero spatial curvature: $k = 0$

In this case, if we set $r = \chi$ (to keep our notation consistent), the 3-space line element is

$$d\sigma^2 = R^2 [d\chi^2 + \chi^2(d\theta^2 + \sin^2 \theta d\phi^2)],$$

which is simply the ordinary three-dimensional Euclidean space. As usual, under the transformation

$$x = R\chi \sin \theta \cos \phi, \quad y = R\chi \sin \theta \sin \phi, \quad z = R\chi \cos \theta,$$

the line element becomes

$$d\sigma^2 = dx^2 + dy^2 + dz^2.$$

Negative spatial curvature: $k = -1$

In this case, it is convenient to introduce a radial coordinate χ given by

$$r = \sinh \chi \quad \Rightarrow \quad dr = \cosh \chi d\chi = (1 + r^2)^{1/2} d\chi,$$

so that the spatial part of the FRW metric becomes

$$d\sigma^2 = R^2 [d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2)].$$

We cannot embed this 3-space in a four-dimensional Euclidean space, but it can be embedded in a four-dimensional Minkowski space with coordinates (w, x, y, z) given by

$$\begin{aligned} w &= R \cosh \chi, \\ x &= R \sinh \chi \sin \theta \cos \phi, \\ y &= R \sinh \chi \sin \theta \sin \phi, \\ z &= R \sinh \chi \cos \theta. \end{aligned}$$

In this case, we can write

$$d\sigma^2 = dw^2 - dx^2 - dy^2 - dz^2,$$

together with the constraint

$$w^2 - x^2 - y^2 - z^2 = R^2,$$

which shows that the 3-space can be represented as a three-dimensional hyperboloid in the four-dimensional Minkowski space. The hypersurface is defined by the coordinate ranges

$$0 \leq \chi \leq \infty, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi \leq 2\pi.$$

The 2-surfaces $\chi = \text{constant}$ are 2-spheres with surface area

$$A = 4\pi R^2 \sinh^2 \chi,$$

which increases indefinitely as χ increases. The proper radius of such a 2-sphere is $R\chi$, and so the surface area is larger than the corresponding result in Euclidean space. The total volume of the space is infinite.

From the above discussion, we see that a convenient form for the FRW metric is

$$ds^2 = c^2 dt^2 - R^2(t) [d\chi^2 + S^2(\chi)(d\theta^2 + \sin^2 \theta d\phi^2)], \quad (14.11)$$

where the function $r = S(\chi)$ is given by

$$S(\chi) = \begin{cases} \sin \chi & \text{if } k = 1, \\ \chi & \text{if } k = 0, \\ \sinh \chi & \text{if } k = -1. \end{cases} \quad (14.12)$$

Once again, it is clear that (χ, θ, ϕ) are *comoving coordinates*.

14.8 Geodesics in the FRW metric

In the comoving coordinate system(s) we have defined above, the galaxies have fixed spatial coordinates (by construction; any peculiar velocities are ignored). Thus the 'cosmological fluid' is at rest in the comoving frame we have chosen. We now consider the motion of particles travelling with respect to this comoving frame. In particular, we consider the geodesic motion of 'free' particles, i.e. those experiencing only the 'background' gravitational field of the cosmological fluid and no other forces. Examples of such particles might include a projectile shot out of a galaxy or a photon travelling through intergalactic space. We could use the 'Lagrangian' procedure to calculate the geodesic equations for the FRW metric, but instead we take advantage of the fact that the spatial part of the metric is homogeneous and isotropic to arrive at the equations rather more quickly.

It is convenient to express the FRW metric in the form (14.11) and write $[x^\mu] = (t, \chi, \theta, \phi)$, so that

$$g_{00} = c^2, \quad g_{11} = -R^2(t), \quad g_{22} = -R^2(t)S^2(\chi), \quad g_{33} = -R^2(t)S^2(\chi)\sin^2\theta.$$

The path of a particle is given by the geodesic equation

$$\dot{u}^\mu + \Gamma^\mu_{\nu\sigma} u^\nu u^\sigma = 0,$$

where $u^\mu = \dot{x}^\mu$ and the dot corresponds to differentiation with respect to some affine parameter. For our present purposes, however, it will be more useful to rewrite the geodesic equation in the form

$$\dot{u}_\mu = \frac{1}{2}(\partial_\mu g_{\nu\sigma})u^\nu u^\sigma,$$

which shows, as expected, that if the metric is independent of a particular coordinate x^λ then u_λ is conserved along the geodesic.

Let us suppose that the geodesic passes through some spatial point P . Since the spatial part of the metric is spatially homogeneous and isotropic we can, without loss of generality, take the spatial origin of the coordinate system, i.e. $\chi = 0$, to be at the point P . This simplifies the analysis considerably.

Consider first the ϕ -component u^3 . Since the metric is independent of ϕ , we have $\dot{u}_3 = 0$ so that u_3 is constant along the geodesic. But

$$u_3 = g_{33}u^3 = -R^2(t)S^2(\chi)\sin^2\theta u^3,$$

so that $u_3 = 0$ at the point P where $\chi = 0$. Thus $u_3 = 0$ along the path and so also we have $u^3 = \dot{\phi} = 0$. Hence, along the geodesic,

$$\phi = \text{constant.}$$

For the θ -component, we have

$$\dot{u}_2 = \frac{1}{2}(\partial_2 g_{\nu\sigma})u^\nu u^\sigma. \quad (14.13)$$

The only component of $g_{\mu\nu}$ that depends on $x^2 = \theta$ is g_{33} , but the contribution of the corresponding term in (14.13) vanishes since $u^3 = 0$. Thus $\dot{u}_2 = 0$ and so u_2 is constant along the geodesic. Again

$$u_2 = g_{22}u^2 = -R^2(t)S^2(\chi)u^2,$$

which vanishes at $P(\chi = 0)$, and so u_2 is zero along the geodesic, as is u^2 , so that

$$\theta = \text{constant.}$$

For the r -component,

$$\dot{u}_1 = \frac{1}{2}(\partial_1 g_{\nu\sigma})u^\nu u^\sigma. \quad (14.14)$$

We have $u^2 = u^3 = 0$, while g_{00} and g_{11} are independent of χ . Thus, $\dot{u}_1 = 0$ so that u_1 is constant along the geodesic, so $u_1 = g_{11}u^1$ must be constant. Thus, we have

$$R^2(t)\dot{\chi} = \text{constant.} \quad (14.15)$$

Finally, u^0 can be found from the appropriate normalisation condition, $u^\mu u_\mu = c^2$ for massive particles or $u^\mu u_\mu = 0$ for photons. Thus, we have

$$\dot{t}^2 = \begin{cases} 1 + \frac{R^2(t)\dot{\chi}^2}{c^2} & \text{for a massive particle,} \\ \frac{R^2(t)\dot{\chi}^2}{c^2} & \text{for a photon.} \end{cases}$$

14.9 The cosmological redshift

We can use the results of the last section to derive the *cosmological redshift*. Suppose that a photon is emitted at cosmic time t_E by a comoving observer with fixed spatial coordinates $(\chi_E, \theta_E, \phi_E)$ and that the photon is received at time t_R by another observer at fixed comoving coordinates. We may take the latter observer to be at the origin of our spatial coordinate system.

For a photon one can choose an affine parameter such that the 4-momentum is $p^\mu = \dot{x}^\mu$. From our above discussion, $d\theta = d\phi = 0$ along the photon geodesic, or equivalently $p_2 = p_3 = 0$, and (14.15) shows that p_1 is constant along the geodesic. Since the photon momentum is null, we also require that $g^{\mu\nu}p_\mu p_\nu = 0$, which reduces to

$$\frac{1}{c^2}(p_0)^2 - \frac{1}{R^2(t)}(p_1)^2 = 0,$$

from which we find $p_0 = cp_1/R(t)$.

In Appendix 9A we showed that, for an emitter and receiver with fixed spatial coordinates, the frequency shift of the photon is given, in general, by

$$\frac{\nu_R}{\nu_E} = \frac{p_0(R)}{p_0(E)} \left[\frac{g_{00}(E)}{g_{00}(R)} \right]^{1/2}. \quad (14.16)$$

For the FRW metric we have $g_{00} = c^2$, and so we find immediately that

$$1 + z \equiv \frac{\nu_E}{\nu_R} = \frac{R(t_R)}{R(t_E)}. \quad (14.17)$$

Thus we see that if the scale factor $R(t)$ is increasing with cosmic time, so that the universe is *expanding*, then the photon is *redshifted* by an amount z . Conversely,

if the universe were contracting then the photon would be blueshifted. Only if the universe were *static*, so that $R = \text{constant}$, would there be no frequency shift.

In fact, we may also arrive at this result directly from the FRW metric. Since $ds = d\theta = d\phi = 0$ along the photon path, from (14.11) we have, for an incoming photon,

$$\int_{t_E}^{t_R} \frac{c dt}{R(t)} = \int_0^{\chi_E} d\chi.$$

Now, if the emitter sends a second light pulse at time $t_E + \delta t_E$, which is received at time $t_R + \delta t_R$, then

$$\int_{t_E + \delta t_E}^{t_R + \delta t_R} \frac{c dt}{R(t)} = \int_0^{\chi_E} d\chi = \int_{t_E}^{t_R} \frac{c dt}{R(t)},$$

from which we see immediately that

$$\int_{t_R}^{t_R + \delta t_R} \frac{c dt}{R(t)} = \int_{t_E}^{t_E + \delta t_E} \frac{c dt}{R(t)}.$$

Assuming that δt_E and δt_R are small, so that $R(t)$ can be taken as constant in both integrals, we have

$$\frac{\delta t_R}{R(t_R)} = \frac{\delta t_E}{R(t_E)}.$$

Considering the pulses to be the successive wavecrests of an electromagnetic wave, we again find that

$$1 + z \equiv \frac{\nu_E}{\nu_R} = \frac{\delta t_R}{\delta t_E} = \frac{R(t_R)}{R(t_E)}.$$

14.10 The Hubble and deceleration parameters

In a common notation we shall write the present cosmic time, or epoch, as t_0 . Thus photons received today from distant galaxies are received at t_0 . If the emitting galaxy is nearby and emits a photon at cosmic time t , we can write $t = t_0 - \delta t$, where $\delta t \ll t_0$. Thus, let us expand the scale factor $R(t)$ as a power series about the present epoch t_0 to obtain

$$\begin{aligned} R(t) &= R[t_0 - (t_0 - t)] \\ &= R(t_0) - (t_0 - t)\dot{R}(t_0) + \frac{1}{2}(t_0 - t)^2\ddot{R}(t_0) - \dots \\ &= R(t_0) \left[1 - (t_0 - t)H(t_0) - \frac{1}{2}(t_0 - t)^2 q(t_0)H^2(t_0) - \dots \right], \quad (14.18) \end{aligned}$$

where we have introduced the *Hubble parameter* $H(t)$ and the *deceleration parameter* $q(t)$. These are given by

$$\boxed{\begin{aligned} H(t) &\equiv \frac{\dot{R}(t)}{R(t)}, \\ q(t) &\equiv -\frac{\ddot{R}(t)R(t)}{\dot{R}^2(t)}, \end{aligned}} \quad (14.19)$$

where the dot corresponds to differentiation with respect to cosmic time t . It should be noted that these definitions are valid at any cosmic time. The present-day values of these parameters are usually denoted by $H_0 \equiv H(t_0)$ and $q_0 \equiv q(t_0)$.

Using these definitions, we can write the redshift z in terms of the 'look-back time' $t - t_0$ as

$$z = \frac{R(t_0)}{R(t)} - 1 = \left[1 - (t_0 - t)H_0 - \frac{1}{2}(t_0 - t)^2 q_0 H_0^2 - \dots \right]^{-1} - 1$$

and, assuming that $t_0 - t \ll t_0$, we have

$$z = (t_0 - t)H_0 + (t_0 - t)^2 \left(1 + \frac{1}{2}q_0 \right) H_0^2 + \dots \quad (14.20)$$

Since it is the redshift that is an observable quantity, it is more useful to invert the above power series to obtain the look-back time $t_0 - t$ in terms of z . Thus for $z \ll 1$ we have

$$\boxed{t_0 - t = H_0^{-1} z - H_0^{-1} \left(1 + \frac{1}{2}q_0 \right) z^2 + \dots} \quad (14.21)$$

It is worth noting that, as one might expect in this approximation, the relations (14.20) and (14.21) depend only on the present-day values H_0 and q_0 of the Hubble and deceleration parameters and hence may be evaluated without knowledge of the complete expansion history $R(t)$ of the universe.

Using the Taylor expansion (14.18), we can also obtain an approximate expression for the χ -coordinate of the emitting galaxy, which is given by

$$\chi = \int_t^{t_0} \frac{c \, dt}{R(t)} = \int_t^{t_0} c R_0^{-1} [1 - (t_0 - t)H_0 - \dots]^{-1} dt.$$

Assuming once more that $t_0 - t \ll t_0$, we have

$$\chi = c R_0^{-1} \left[(t_0 - t) + \frac{1}{2}(t_0 - t)^2 H_0 + \dots \right]. \quad (14.22)$$

We may now substitute for the look-back time $t_0 - t$ in this result using (14.21), to obtain an expression for the χ -coordinate of the emitting galaxy in terms of its redshift (assuming $z \ll 1$), which reads

$$\chi = \frac{c}{R_0 H_0} \left[z - \frac{1}{2}(1 + q_0)z^2 + \cdots \right]. \quad (14.23)$$

Once again, in this approximation the results (14.22) and (14.23) only depend on the present-day values H_0 and q_0 and may be evaluated without knowing the expansion history of the universe.

From the FRW metric, we see that the proper distance d to the emitting galaxy³ at cosmic time t_0 is $d = R_0 \chi$. Thus, for very nearby galaxies, $d \approx c(t_0 - t)$. Moreover, from (14.20), in this case $z \approx (t_0 - t)H_0$. So, if we were to interpret the cosmological redshift as a Doppler shift due to a recession velocity v of the emitting galaxy, we would obtain

$$v = cz = H_0 d, \quad (14.24)$$

which is approximately valid for small z . The galaxies will therefore appear to recede from us with a recession speed proportional to their distance from us. This is, of course, *Hubble's law*, named after Edwin Hubble, who discovered the expansion of the universe in 1929 by comparing redshifts with distance measurements to nearby galaxies (derived from the period–luminosity relation of Cepheid variables). His results suggested a linear recession law, as in (14.24). This was an amazing result. It implies that the universe started off at high density at some finite time in the past. You will notice from (14.24) that the Hubble ‘constant’ has the dimensions of inverse time. As we will see later, the quantity $1/H_0$ gives the *age of the universe* to within a factor of order unity. It is clear that, in general, the Hubble parameter will vary with cosmic time t and hence with redshift z . By combining the expressions (14.18), (14.19) and (14.21), we can obtain an expression for how the Hubble parameter varies with z for small redshift,

$$H(z) = H_0 [1 + (1 + q_0)z - \cdots]. \quad (14.25)$$

So far, we have been considering the low- z limit. Having introduced the Hubble parameter, however, we may use it to derive useful *general* expressions for the

³ In order to measure the proper distance d , one would in fact have to arrange for all the ‘civilisations’ along the route to the galaxy to lay out measuring rods at the same cosmic time t_0 . This could be synchronised by, for example, requiring the temperature of the cosmic microwave background or the mean matter density of the universe to have a given value. We will discuss more practical measures of distance shortly.

look-back time to an emitting galaxy, and for its χ -coordinate, as functions of the redshift z of the received photon. In general, we have

$$dz = d(1+z) = d\left(\frac{R_0}{R}\right) = -\frac{R_0}{R} \dot{R} dt = -(1+z)H(z) dt,$$

which provides a very useful relation between an interval dz in redshift and the corresponding interval dt in cosmic time. Thus, we can write the look-back time as

$$t - t_0 = \int_t^{t_0} dt = \int_0^z \frac{dz}{(1+z)H(z)}, \quad (14.26)$$

and the galaxy's χ -coordinate is given by

$$\chi = \int_t^{t_0} \frac{c dt}{R(t)} = \frac{c}{R_0} \int_0^z \frac{dz}{H(z)}. \quad (14.27)$$

It is clear, however, that in order to evaluate either of these integrals we must know how $H(z)$ varies with z , which requires knowledge of the evolution of the scale factor $R(t)$.

14.11 Distances in the FRW geometry

Distance measures in an expanding universe can be confusing. For example, let us consider the distance to some remote galaxy. The light received from the galaxy was emitted when the universe was younger, because light travels at a finite speed c . Evidently, as we look at more distant objects, we see them as they were at an earlier time in the universe's history *when proper distances were smaller*, since the universe is expanding. What, therefore, do we mean by the 'distance' to a galaxy? In fact, interpreting and calculating distances in an expanding universe is straightforward, but one must be clear about what is meant by 'distance'.

From the FRW metric

$$ds^2 = c^2 dt^2 - R^2(t) [d\chi^2 + S^2(\chi)(d\theta^2 + \sin^2 \theta d\phi^2)],$$

we can define a number of different measures of distance. The parameter χ is a comoving coordinate that is sometimes referred to as the *coordinate distance*, whereas the *proper distance* to an object at some cosmic time t is $d = R(t)\chi$, but this cannot be measured in practice. Thus, we must look for alternative ways of defining the distance to an object. The two most important *operationally defined* distance measures are the *luminosity distance* and the *angular diameter distance*. These distance measures form the basis for observational tests of the geometry of the universe.

Luminosity distance

In an ordinary static Euclidean universe, if a source of *absolute luminosity* L (measured in $\text{W} = \text{J s}^{-1}$) is at a distance d then the *flux* that we receive (measured in W m^{-2}) is $F = L/(4\pi d^2)$. Now suppose that we are actually in an expanding FRW geometry, but we know that the source has a luminosity L and we observe a flux F . The quantity

$$d_L = \left(\frac{L}{4\pi F} \right)^{1/2}, \quad (14.28)$$

is called the *luminosity distance* of the source. This is an *operational definition*, and we must now investigate how to express it in terms of the FRW metric.

Consider an emitting source E with a fixed comoving coordinate χ relative to an observer O (note that, by symmetry, the emitter would assign the *same* value of χ to the observer). We assume that the absolute luminosity of E as a function of cosmic time is $L(t)$ and that the photons it emits are detected by O at cosmic time t_0 . Clearly, the photons must have been emitted at an earlier time t_e . Assuming the photons to have been emitted isotropically, the radiation will be spread evenly over a sphere centred on E and passing through O (see Figure 14.2). The proper area of this sphere is

$$A = 4\pi R^2(t_0)S^2(\chi).$$

However, each photon received by O is redshifted in frequency, so that

$$\nu_0 = \frac{\nu_e}{1+z},$$

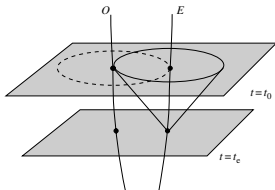


Figure 14.2 Geometry associated with the definition of luminosity distance (with one spatial dimension suppressed).

and, moreover, the arrival rate of the photons is also reduced by the same factor. Thus, the *observed* flux at O is

$$F(t_0) = \frac{L(t_e)}{4\pi[R_0S(\chi)]^2} \frac{1}{(1+z)^2}.$$

The luminosity distance defined above is now evaluated as

$$d_L = R_0S(\chi)(1+z). \quad (14.29)$$

This is an important quantity, which can be used practically, but note that it depends on the time history of the scale factor through the dependence on χ .

Angular diameter distance

Another important distance measure is based upon the notion of the existence of some standard-length ‘rods’, whose angular diameter we can observe. Suppose that a source has proper diameter ℓ . Then, in Euclidean space, if it were at a distance d it would subtend an angular diameter $\Delta\theta = \ell/d$. In an FRW geometry, we thus define the *angular diameter distance* to an object to be

$$d_A = \frac{\ell}{\Delta\theta}. \quad (14.30)$$

This is again an operational definition, and we now investigate how to express it in terms of the FRW metric.

Suppose we have two radial null geodesics (light paths) meeting at the observer at time t_0 with an angular separation $\Delta\theta$, having been emitted at time t_e from a source of proper diameter ℓ at a fixed comoving coordinate χ (assuming, for simplicity, that the spatial axes are oriented so that $\phi = \text{constant}$ along the photon paths); see Figure 14.3. To obtain a clearer view of the specification of the coordinates, we may look vertically down the worldline of O and define the coordinates as in Figure 14.4. From the angular part of the FRW metric we have

$$\ell = R(t_e)S(\chi)\Delta\theta,$$

so that

$$d_A = R(t_e)S(\chi) = R(t_0) \frac{R(t_e)}{R(t_0)} S(\chi) = \frac{R(t_0)S(\chi)}{1+z}.$$

Thus the angular diameter distance is given by

$$d_A = \frac{R_0S(\chi)}{1+z}. \quad (14.31)$$

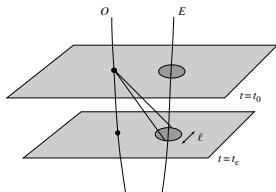


Figure 14.3 Geometry associated with the definition of angular diameter distance (with one spatial dimension suppressed).

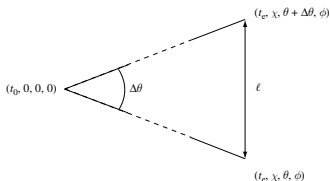


Figure 14.4 Specification of the coordinates in the definition of angular diameter distance.

This differs from the luminosity distance d_L by a factor $(1+z)^2$, emphasizing again that ‘distance’ depends on definition. Again, because of the χ -dependence we need to know the time history of the scale factor $R(t)$ to evaluate d_A .

14.12 Volumes and number densities in the FRW geometry

The interpretation of cosmological observations often requires one to determine the volume of some three-dimensional region of the FRW geometry. Consider a comoving cosmological observer, whom we may take to be at the origin $\chi = 0$ of our comoving coordinate system. From the FRW metric

$$ds^2 = c^2 dt^2 - R^2(t) [d\chi^2 + S^2(\chi)(d\theta^2 + \sin^2 \theta d\phi^2)],$$

we see that, at cosmic time t_0 , the proper volume of the region of space lying in the infinitesimal coordinate range $\chi \rightarrow \chi + d\chi$ and subtending an infinitesimal solid angle $d\Omega = \sin \theta d\theta d\phi$ at the observer is

$$dV_0 = (R_0 d\chi) [R_0^2 S^2(\chi) d\Omega] = R_0^3 S^2(\chi) d\chi d\Omega.$$

For the interval $\chi \rightarrow \chi + d\chi$ in the radial comoving coordinate there exists a corresponding interval $z \rightarrow z + dz$ in the redshift of objects lying in this radial range (and also a corresponding cosmic time interval $t \rightarrow t + dt$ within which the light observed by O at $t = t_0$ was emitted). We may therefore write the volume element as

$$dV_0 = R_0^3 S^2(\chi) \frac{d\chi}{dz} dz d\Omega.$$

From (14.27), however, we have

$$\frac{d\chi}{dz} = \frac{c}{R_0 H(z)},$$

and so

$$dV_0 = \frac{c R_0^3 S^2(\chi(z))}{H(z)} dz d\Omega,$$

where we have made explicit that χ is also a function of z . This volume element is illustrated in Figure 14.5. For an expanding universe, the proper volume of this

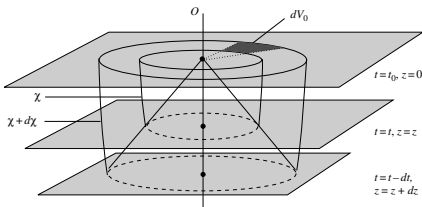


Figure 14.5 Geometry associated with the definition of a proper volume element dV_0 at cosmic time $t = t_0$ (with one spatial dimension suppressed).

comoving region will be smaller at some earlier cosmic time t (which corresponds to some redshift z). Indeed, using (14.17), we have

$$dV(z) = \frac{dV_0}{(1+z)^3} = \frac{cR_0^2 S^2(\chi(z))}{(1+z)^3 H(z)} dz d\Omega. \quad (14.32)$$

The main use of the result (14.32) is in predicting the number of galaxies (of a certain type) that one would expect to observe in a given area of sky and redshift interval, and comparing that result with observations. Suppose, for example, that the proper number density of galaxies of a certain type at a redshift z is given by $n(z)$. Using (14.32), the total number dN of such objects in the redshift interval $z \rightarrow z + dz$ and in a solid angle $d\Omega$ is

$$dN = n(z) dV(z) = \frac{cR_0^2 S^2(\chi(z))}{H(z)} \frac{n(z)}{(1+z)^3} dz d\Omega. \quad (14.33)$$

The above expression has been arranged to make use of the fact that, if objects are conserved (so that, once formed, galaxies are not later destroyed), we may write $n(z)/(1+z)^3 = n_0$, where n_0 is the present-day proper number density of such objects; hence the resulting expression is simplified somewhat. As an illustration, let us consider a population of galaxies which are formed instantaneously at a redshift $z = z_f$, which are not later destroyed and which have a present-day number density n_0 . From (14.33), the total number of such objects in the whole sky is

$$N = 4\pi c n_0 R_0^2 \int_0^{z_f} \frac{S^2(\chi(z))}{H(z)} dz.$$

Clearly, in order to evaluate this integral one requires knowledge of the expansion history $R(t)$ of the universe.

14.13 The cosmological field equations

So far we have investigated only the geometric and kinematic consequences of the FRW metric. The *dynamics* of the spacetime geometry is characterised entirely by the scale factor $R(t)$. In order to determine the function $R(t)$, we must solve the gravitational field equations in the presence of matter.

From Chapter 8 the gravitational field equations, in the presence of a non-zero cosmological constant, are

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = -\kappa T_{\mu\nu},$$

where $\kappa = 8\pi G/c^4$. It is, however, more convenient to express the field equations in the alternative form

$$R_{\mu\nu} = -\kappa(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}) + \Lambda g_{\mu\nu}, \quad (14.34)$$

where $T = T^\mu_\mu$. In order to solve these equations, we clearly need a model for the energy-momentum tensor of the matter that fills the universe. For simplicity, we shall grossly idealise the universe and model the matter by a simple macroscopic fluid, devoid of shear-viscous, bulk-viscous and heat-conductive properties. Thus we assume a *perfect fluid*, which is characterised at each point by its proper density ρ and the pressure p in the instantaneous rest frame. The energy-momentum tensor is given by

$$T^{\mu\nu} = \left(\rho + \frac{p}{c^2} \right) u^\mu u^\nu - p g^{\mu\nu}. \quad (14.35)$$

Since we are seeking solutions for a homogeneous and isotropic universe, the density ρ and pressure p must be functions of cosmic time t alone.

We may perform the calculation in any coordinate system, but the algebra is simplified slightly by adopting the comoving coordinates $[x^\mu] = (t, r, \theta, \phi)$, in which the FRW metric takes the form

$$ds^2 = c^2 dt^2 - R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right].$$

Thus the covariant components $g_{\mu\nu}$ of the metric are

$$g_{00} = c^2, \quad g_{11} = -\frac{R^2(t)}{1 - kr^2}, \quad g_{22} = -R^2(t)r^2, \quad g_{33} = -R^2(t)r^2 \sin^2 \theta.$$

Since the metric is diagonal, the contravariant components $g^{\mu\nu}$ are simply the reciprocals of the covariant components.

The connection is given in terms of the metric by

$$\Gamma^\sigma_{\mu\nu} = \frac{1}{2} g^{\sigma\rho} (\partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu}),$$

from which it is straightforward to show that the only non-zero coefficients are

$$\begin{aligned} \Gamma^0_{11} &= R\dot{R}/[c(1 - kr^2)], & \Gamma^0_{22} &= R\dot{R}r^2/c, & \Gamma^0_{33} &= (R\dot{R}r^2 \sin^2 \theta)/c, \\ \Gamma^1_{01} &= c\dot{R}/R, & \Gamma^1_{11} &= kr/(1 - kr^2), & \Gamma^0_{33} &= (R\dot{R}r^2 \sin^2 \theta)/c, \\ \Gamma^1_{33} &= -r(1 - kr^2) \sin^2 \theta, \\ \Gamma^2_{02} &= c\dot{R}/R, & \Gamma^2_{12} &= 1/r, & \Gamma^2_{33} &= \sin \theta \cos \theta, \\ \Gamma^3_{03} &= c\dot{R}/R, & \Gamma^3_{13} &= 1/r, & \Gamma^3_{23} &= \cot \theta, \end{aligned}$$

where the dots denote differentiation with respect to cosmic time t . We next substitute these expressions for the connection coefficients into the expression for the Ricci tensor,

$$R_{\mu\nu} = \partial_\nu \Gamma^\sigma_{\mu\sigma} - \partial_\sigma \Gamma^\sigma_{\mu\nu} + \Gamma^\rho_{\mu\sigma} \Gamma^\sigma_{\rho\nu} - \Gamma^\rho_{\mu\nu} \Gamma^\sigma_{\rho\sigma}.$$

After some tedious but straightforward algebra, we find that the off-diagonal components of the Ricci tensor are zero and the diagonal components are given by

$$\begin{aligned} R_{00} &= 3\ddot{R}/R, \\ R_{11} &= -(R\ddot{R} + 2\dot{R}^2 + 2c^2k)c^{-2}/(1 - kr^2), \\ R_{22} &= -(R\ddot{R} + 2\dot{R}^2 + 2c^2k)c^{-2}r^2, \\ R_{33} &= -(R\ddot{R} + 2\dot{R}^2 + 2c^2k)c^{-2}r^2 \sin^2 \theta. \end{aligned}$$

We must now turn our attention to the right-hand side of the field equations (14.34). In our comoving coordinate system (t, r, θ, ϕ) , the 4-velocity of the fluid is simply

$$[u^\mu] = (1, 0, 0, 0),$$

which we can write as $u^\mu = \delta_0^\mu$. Thus the covariant components of the 4-velocity are

$$u_\mu = g_{\mu\nu}\delta_0^\nu = g_{\mu 0} = c^2\delta_\mu^0,$$

so we can write the energy–momentum tensor (14.35) as

$$T_{\mu\nu} = (\rho c^2 + p)c^2\delta_\mu^0\delta_\nu^0 - pg_{\mu\nu}.$$

Moreover, since $u^\mu u_\mu = c^2$, contraction of the energy–momentum tensor gives

$$T = T^\mu_\mu = \left(\rho + \frac{p}{c^2}\right)c^2 - p\delta^\mu_\mu = \rho c^2 - 3p.$$

Hence we can write the terms on the right-hand side of the field equations (14.34) that depend on the energy–momentum as

$$T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu} = (\rho c^2 + p)c^2\delta_\mu^0\delta_\nu^0 - \frac{1}{2}(\rho c^2 - p)g_{\mu\nu}.$$

Including the cosmological-constant term, we find that the right-hand side of the field equations (14.34) vanishes for $\mu \neq \nu$. The non-zero components read

$$\begin{aligned} -\kappa(T_{00} - \tfrac{1}{2}Tg_{00}) + \Lambda g_{00} &= -\tfrac{1}{2}\kappa(\rho c^2 + 3p)c^2 + \Lambda c^2, \\ -\kappa(T_{11} - \tfrac{1}{2}Tg_{11}) + \Lambda g_{11} &= -\left[\tfrac{1}{2}\kappa(\rho c^2 - p) + \Lambda\right]R^2/(1 - kr^2), \\ -\kappa(T_{22} - \tfrac{1}{2}Tg_{22}) + \Lambda g_{22} &= -\left[\tfrac{1}{2}\kappa(\rho c^2 - p) + \Lambda\right]R^2r^2, \\ -\kappa(T_{33} - \tfrac{1}{2}Tg_{33}) + \Lambda g_{33} &= -\left[\tfrac{1}{2}\kappa(\rho c^2 - p) + \Lambda\right]R^2r^2 \sin^2 \theta. \end{aligned}$$

Combining these expressions with those for the components of the Ricci tensor, we see that the three spatial field equations are equivalent, which is essentially

due to the homogeneity and isotropy of the FRW metric. Thus the gravitational field equations yield just the two independent equations,

$$\begin{aligned} 3\ddot{R}/R &= -\frac{1}{2}\kappa(\rho c^2 + 3p)c^2 + \Lambda c^2, \\ R\ddot{R} + 2\dot{R}^2 + 2c^2k &= \left[\frac{1}{2}\kappa(\rho c^2 - p) + \Lambda\right]c^2 R^2. \end{aligned}$$

Eliminating $\ddot{R}$ from the second equation and remembering that $\kappa = 8\pi G/c^4$, we finally arrive at the *cosmological field equations*

$$\begin{aligned} \ddot{R} &= -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right)R + \frac{1}{3}\Lambda c^2 R, \\ \dot{R}^2 &= \frac{8\pi G}{3}\rho R^2 + \frac{1}{3}\Lambda c^2 R^2 - c^2 k. \end{aligned} \tag{14.36}$$

These two differential equations determine the time evolution of the scale factor $R(t)$ and are known as the *Friedmann–Lemaître equations*. In the case $\Lambda = 0$ they are often called simply the *Friedmann equations*. We will discuss the solutions to these equations in various cases in Chapter 15.

14.14 Equation of motion for the cosmological fluid

For any particular model of the universe, the two cosmological field equations (14.36) are sufficient to determine $R(t)$. Nevertheless, we can derive one further important equation (which is often useful in shortening calculations) from the fact that energy–momentum conservation requires

$$\nabla_\mu T^{\mu\nu} = 0.$$

From our discussion of a perfect fluid in Chapter 8, we know that this requirement leads to the relativistic equations of continuity and motion for the cosmological fluid. These equations read

$$\nabla_\mu(\rho u^\mu) + \frac{p}{c^2}\nabla_\mu u^\mu = 0, \tag{14.37}$$

$$\left(\rho + \frac{p}{c^2}\right)u^\mu \nabla_\mu u^\nu = \left(g^{\mu\nu} - \frac{u^\mu u^\nu}{c^2}\right)\nabla_\mu p. \tag{14.38}$$

The second equation is easily shown to be satisfied identically, since both sides equal zero. This confirms that the fluid particles (galaxies) follow geodesics, which was to be expected since p is a function of t alone, and so there is no pressure gradient to push them off geodesics. The continuity equation (14.37) can be written

$$(\partial_\mu p)u^\mu + \left(\rho + \frac{p}{c^2}\right)(\partial_\mu u^\mu + \Gamma^\mu_{\nu\mu}u^\nu) = 0.$$

Remembering that ρ is a function of t alone, and with $u^\mu = \delta_0^\mu$, this reduces to

$$\dot{\rho} + \left(\rho + \frac{p}{c^2}\right) \frac{3\dot{R}}{R} = 0, \quad (14.39)$$

which expresses energy conservation. This equation can in fact be derived directly from the field equations (14.36) by eliminating $\ddot{R}$. Thus, only two of the three equations (14.36) and (14.39) are independent. One may use whichever two equations are most convenient in any particular calculation.

Equation (14.39) can be simply rearranged into the useful alternative form

$$\frac{d(\rho R^3)}{dt} = -\frac{3p\dot{R}R^2}{c^2}. \quad (14.40)$$

Moreover, by transforming the derivative with respect to t to a derivative with respect to R , one obtains a third useful form of the equation, namely

$$\frac{d(\rho R^3)}{dR} = -\frac{3pR^2}{c^2}. \quad (14.41)$$

Finally, we note that the density and pressure of a fluid are related by its equation of state. In cosmology, it is usual to assume that (each component of) the cosmological fluid has an equation of state of the form

$$p = w\rho c^2,$$

where the *equation-of-state parameter* w is a constant (in the more exotic cosmological models one sometimes allows w to be a function of cosmic time t , but we shall not consider such models here). The energy equation (14.41) can then be written

$$\frac{d(\rho R^3)}{dR} = -3w\rho R^2.$$

This equation has the immediate solution

$$\rho \propto R^{-3(1+w)}, \quad (14.42)$$

which gives the evolution of the density ρ as a function of the scale factor $R(t)$. Note that in general ρc^2 is the energy density of the fluid. In particular $w = 0$ for pressureless ‘dust’, $w = \frac{1}{3}$ for radiation and $w = -1$ for the vacuum (if the cosmological constant $\Lambda \neq 0$; see Section 8.7).

14.15 Multiple-component cosmological fluid

Suppose that the cosmological fluid in fact consists of several distinct components (for example, matter, radiation and the vacuum) that do not interact except through their mutual gravitation. Let us suppose further that each component can be modelled as a perfect fluid, as discussed above.

The energy–momentum tensor of a multiple-component fluid is given simply by

$$T^{\mu\nu} = \sum_i (T^{\mu\nu})_i,$$

where i labels the various fluid components. Since each component is modelled as a perfect fluid, we have

$$\begin{aligned} T^{\mu\nu} &= \sum_i \left[\left(\rho_i + \frac{p_i}{c} \right) u^\mu u^\nu - p_i g^{\mu\nu} \right] \\ &= \sum_i \left(\rho_i + \frac{p_i}{c} \right) u^\mu u^\nu - \left(\sum_i p_i \right) g^{\mu\nu}. \end{aligned}$$

Thus, the multicomponent fluid can itself be modelled as a single perfect fluid with

$$\boxed{\rho = \sum_i \rho_i \quad \text{and} \quad p = \sum_i p_i,} \quad (14.43)$$

which can be substituted directly into our cosmological field equations (14.36).⁴

Moreover, since we are assuming that the fluid components are non-interacting, conservation of energy and momentum requires that the condition

$$\nabla_\mu (T^{\mu\nu})_i = 0$$

holds separately for each component. Then each fluid will obey an energy equation of the form (14.39). Thus, if $w_i = p_i/(\rho_i c^2)$ then the density of each fluid evolves independently of the other components as

$$\boxed{\rho_i \propto R^{-3(1+w_i)}}. \quad (14.44)$$

Exercises

14.1 In an N -dimensional manifold, consider the tensor

$$R_{ijkl} = K(g_{ik}g_{jl} - g_{il}g_{jk}),$$

where K may be a function of position. Show that this tensor satisfies the symmetry properties and the cyclic identity of the curvature tensor. Show that, in order to satisfy the Bianchi identity, one requires K to be constant if $N > 2$.

⁴ Unfortunately, if the individual equation-of-state parameters w_i are constants one cannot, in general, define a single effective equation-of-state parameter $w = p/(\rho c^2)$ that is also independent of cosmic time t .

- 14.2 For a 3-space with a line element of the form

$$d\sigma^2 = B(r) dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2,$$

show that the non-zero components of the Ricci tensor are

$$R_{rr} = -\frac{1}{rB} \frac{dB}{dr}, \quad R_{\theta\theta} = \frac{1}{B} - 1 - \frac{r}{2B^2} \frac{dB}{dr}, \quad R_{\phi\phi} = R_{\theta\theta} \sin^2 \theta.$$

Hence show that if the 3-space is maximally symmetric then $B(r)$ must take the form

$$B(r) = \frac{1}{A - Kr^2},$$

where A and K are constants.

- 14.3 In a four-dimensional Euclidean space with ‘Cartesian’ coordinates (w, x, y, z) , a 3-sphere of radius R is defined by $w^2 + x^2 + y^2 + z^2 = R^2$. Show that the metric on the surface of the 3-sphere can be written in the form

$$d\sigma^2 = R^2 [d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2)].$$

Show that the total volume of the 3-sphere is $V = 2\pi^2 R^3$.

- 14.4 In a four-dimensional Minkowski space with ‘Cartesian’ coordinates (w, x, y, z) , a 3-hyperboloid is defined by $w^2 - x^2 - y^2 - z^2 = R^2$. Show that the metric on the surface of the 3-hyperboloid can be written in the form

$$d\sigma^2 = R^2 [d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2)].$$

Show that the total volume of the 3-hyperboloid is infinite.

- 14.5 At cosmic time t_1 , a massive particle is shot out into an expanding FRW universe with velocity v_1 relative to comoving cosmological observers. At a later cosmic time t_2 the particle has a velocity v_2 with respect to comoving cosmological observers. Show that, at any intermediate cosmic time t , the velocity of the particle as measured by a comoving cosmological observer is

$$v(t) = R(t) \frac{d\chi}{dt}.$$

Hence show that

$$\frac{\gamma_{v_2} v_2}{\gamma_{v_1} v_1} = \frac{R(t_1)}{R(t_2)},$$

where $\gamma_v = (1 - v^2/c^2)^{-1/2}$ and $R(t)$ is the scale factor at cosmic time t . By considering the particle momentum, show that as $v_1 \rightarrow c$ the photon redshift formula is recovered.

- 14.6 In the limit $z \ll 1$, show that the look-back time for a galaxy with redshift z is

$$t_0 - t = H_0^{-1} z - H_0^{-1} \left(1 + \frac{1}{2} q_0\right) z^2 + \dots$$

Show also that, in this limit, the variation of the Hubble parameter with redshift is given by

$$H(z) = H_0[1 + (1 + q_0)z - \dots].$$

- 14.7 In a spatially flat FRW geometry, show that the luminosity and angular diameter distances to an object of redshift z are given, in the limit $z \ll 1$, by

$$d_L = \frac{c}{H_0} \left[z + \frac{1}{2}(1 - q_0)z^2 + \dots \right],$$

$$d_A = \frac{c}{H_0} \left[z - \frac{1}{2}(3 + q_0)z^2 + \dots \right].$$

Hence show that the angular diameter of a standard object can increase as z increases. Do these results still hold in a spatially curved FRW geometry?

- 14.8 In the FRW geometry, show that the look-back time to a nearby object at proper distance d is

$$t_0 - t = \frac{d}{c} - \frac{H_0 d^2}{2c^2} + \dots$$

Hence show that the redshift to the object is

$$z = \frac{H_0 d}{c} + \frac{1 + q_0}{2} \frac{H_0^2 d^2}{c^2} + \dots$$

- 14.9 The observed flux in the frequency range $[\nu_1, \nu_2]$ received from some distant comoving object is given by

$$F_{\text{obs}}(\nu_1, \nu_2) = \int_{\nu_1}^{\nu_2} f_{\text{obs}}(\nu) d\nu,$$

where $f_{\text{obs}}(\nu)$ is the observed flux density (in $\text{W m}^{-2} \text{Hz}^{-1}$) as a function of frequency. If $f_{\text{em}}(\nu)$ is the emitted (or intrinsic) flux density of the object, show that

$$f_{\text{obs}}(\nu) = \frac{f_{\text{em}}((1+z)\nu)}{1+z},$$

where z is the redshift of the object. If $f_{\text{em}}(\nu) \propto \nu^\alpha$ over a wide range of frequencies, show that

$$F_{\text{obs}}(\nu_1, \nu_2) = K_z F_{\text{em}}(\nu_1, \nu_2),$$

where the *K-correction* is given by $K_z = (1+z)^{\alpha-1}$.

- 14.10 The observed surface brightness Σ_{obs} of an extended object observed in the frequency range $[\nu_1, \nu_2]$ is defined as the observed flux per unit solid angle. Thus, for a (small) circular object subtending an angular diameter $\Delta\theta$ we have

$$\Sigma_{\text{obs}} = \frac{4F_{\text{obs}}(\nu_1, \nu_2)}{\pi(\Delta\theta)^2},$$

where $F_{\text{obs}}(\nu_1, \nu_2)$ is defined in Exercise 14.9. Show that Σ_{obs} can be written as

$$\Sigma_{\text{obs}} = \frac{4}{\pi} \frac{L_{\text{em}}(\nu_1, \nu_2)}{4\pi\ell^2} \frac{K_z}{(1+z)^4},$$

where ℓ is the physical (projected) diameter of the object, $L_{\text{em}}(\nu_1, \nu_2)$ is the intrinsic luminosity of the object in the frequency range $[\nu_1, \nu_2]$ and K_z is the K -correction.

Note: The above result is independent of cosmological parameters. Moreover, setting aside the K -correction, the $(1+z)^{-4}$ -dependence means that the surface brightness of extended objects drops very rapidly with redshift, making the detection of high- z objects difficult.

- 14.11 A commonly used distance measure in cosmology is the proper-motion distance d_M defined by

$$d_M = \frac{v}{\dot{\theta}},$$

where v is the proper transverse velocity of (some part of) the object, which is assumed known from astrophysics, and $\dot{\theta}$ is the corresponding observed angular velocity. Show that

$$d_M = (1+z)d_A = \frac{d_L}{(1+z)},$$

where d_A and d_L are the angular-diameter distance and the luminosity distance to the object respectively.

- 14.12 A certain population of galaxies undergoes a short ultra-luminous phase at redshift $z = z_*$ that lasts for a proper time interval Δt . After this phase, such galaxies are neither created or destroyed. If $z_* \ll 1$, show that for a spatially flat universe the total number of such galaxies in the sky that are in this phase is given by

$$N = \frac{4\pi c^3 n_0 \Delta t}{H_0^2} \left[z_* + \frac{1}{2}(1 - q_0)z_*^2 + \dots \right],$$

where n_0 is the present-day proper number density of these galaxies.

- 14.13 In the comoving coordinates $[x^\mu] = (t, r, \theta, \phi)$, the FRW metric takes the form

$$ds^2 = c^2 dt^2 - R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right].$$

Using the ‘Lagrangian’ method, or otherwise, calculate the corresponding connection coefficients $\Gamma^\sigma_{\mu\nu}$. Hence calculate the non-zero elements of the Ricci tensor $R_{\mu\nu}$.

- 14.14 In Newtonian cosmology, the universe is modelled as an infinite gas of density $\rho(t)$ that is expanding in such a way that the relative recessional velocity of any two gas particles is $v(t) = H(t)R(t)$, where $H(t) = \dot{R}(t)/R(t)$ and $R(t)$ is the separation

of the particles at time t . Use Gauss' law to determine the force on a particle of mass m on the edge of an arbitrary spherical region, and hence show that

$$\ddot{R} = -\frac{4\pi G}{3}\rho R.$$

By considering the total energy E of the above particle, show further that

$$\dot{R}^2 = \frac{8\pi G}{3}\rho R^2 - c^2 k,$$

where the constant $k = -2E/(mc^2)$. Compare these Newtonian cosmological field equations with their general-relativistic counterparts.

- 14.15 Show that the relativistic equation of motion for the cosmological fluid is satisfied identically and that the relativistic equation of continuity takes the form

$$\dot{\rho} + \left(\rho + \frac{p}{c^2}\right) \frac{3\dot{R}}{R} = 0.$$

Show further that this equation may also be written in the forms

$$\frac{d(\rho R^3)}{dt} = -\frac{3p\dot{R}R^2}{c^2} \quad \text{and} \quad \frac{d(\rho R^3)}{dR} = -\frac{3pR^2}{c^2}.$$

- 14.16 Use the cosmological field equations directly to derive the relativistic equation of continuity for the cosmological fluid given in Exercise 14.15.
- 14.17 Consider a spherical comoving volume of the cosmological fluid whose surface is defined by $\chi = \text{constant}$. As the universe expands show that, for the infinitesimal time interval $t \rightarrow t + dt$, the conservation of energy requires that

$$c^2 \rho V = c^2(\rho + d\rho)(V + dV) + p dV,$$

where ρc^2 and p are the energy density and pressure of the fluid respectively. Hence show that

$$\frac{d\rho}{dR} = -3(1+w)\frac{\rho}{R},$$

where $w = p/(\rho c^2)$ and $R(t)$ is the scale factor of the universe. Show that this equation has the solution $\rho \propto R^{-3(1+w)}$.